Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/06/2018
PI Name:	Riascos-Castaneda, Roy M.D.		
Project Title:	Association of Diffusion Tensor Imaging Parameters of Optic Tracts and Cerebral White Matter Tracts with Visual Impairment and Structural Changes of the Eyes and Optic Nerves in Long-Duration Microgravity Exposure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	9
Comments:	NOTE: Formerly at University of Texas Medical Branch, Galveston, until January 2015.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	02/14/2015	**End Date:**	09/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Allcorn, Aaron	**Contact Phone:**	281.244.8402
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2017 per A. Allcorn/JSC (Ed., 4/1/17) NOTE: Extended to 3/31/2017 per A. Allcorn/JSC (Ed., 9/20/16) NOTE: Extended to 9/30/2016 from 2/13/2016, per NSSC information (Ed., 3/4/16)		
Key Personnel Changes/Previous PI:	January 2018 report: Noam Alperin and Yael Barr are removed from personnel list as most of the work was done in the University of Texas Health Science Center at Houston.		
COI Name (Institution):	Hakimelahi, Reza M.D. (Memorial Sloan Kettering Cancer Center at New York City) Hasan, Khader M Ph.D. (University of Texas Health Science Center at Houston) Kramer, Larry A M.D. (University of Texas Health Science Center at Houston) Sargsyan, Ashot M.D. (Wyle Laboratories, Inc.) Kamali, Arash M.D. (University of Texas Health Science Center at Houston)		
Grant/Contract No.:	NNX15AE25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of grant NNX13AH23G with same title per A. Allcorn/JSC (Ed., 4/1/17), due to Principal Investigator (PI) move to University of Texas Health Science Center, Houston from University of Texas Medical Branch, Galveston. Some crewmembers on long-duration missions have experienced visual impairment and concomitant anatomical changes in the eye detected by ophthalmoscopy and diagnostic imaging. Some of these changes have been reversible and others have persisted for months or even years, and have included increased optic nerve sheath diameter, tortuosity of the optic nerve, and flattening of the posterior globe. The optic nerves are direct extension of the encephalon; hence, some of the changes in the eyes and optic nerves may represent changes occurring in the brain. In the terrestrial environment, similar findings have been described in cases of intracranial hypertension. Subtle changes in the intracranial pressure (ICP) can remain undetected if only anatomical-based imaging is used. Advanced MRI (magnetic resonance imaging) technology, often termed functional MRI, allows seeing beyond the gross anatomy to characterize brain pathways and function; these include tools with the ability to detect and image the motion of fluid around the brain tracts, known as Diffusion Tensor Imaging (DTI). DTI is based on the non isotropic (anisotropic) diffusion of water molecules if exposed to large magnetic fields; water molecules diffuse favoring one direction over the others. Through computer algorithms, a "diffusion tensor" is calculated for each volume element (voxel), allowing to identify voxels representing a given nerve fiber tract. One of the advantages of this system is that it can also detect the magnitude of decrease in signal in the anisotropic direction and calculate the fraction of anisotropy (FA) for a tract. Recently, DTI has been used to evaluate white matter alteration in various brain pathologies, including those with elevated ICP. The corticospinal tract (CST) is the pathway most frequently evaluated by DTI in different conditions, and is especially useful in hydrocephalus. For instance, Hattori et al. (Hattori T, Yuasa T, Aoki S, Sato R, Sawaura H, Mori T, Mizusawa H. Altered microstructure in corticospinal tract in idiopathic normal pressure hydrocephalus: comparison with Alzheimer disease and Parkinson disease with dementia. AJNR Am J Neuroradiol. 2011 Oct;32(9):1681-7.) have shown that patients with idiopathic normal-pressure hydrocephalus (iNPH) have altered microstructures in the CST detected by DTI. Optic nerve tracts have also been studied with DTI for a variety of pathologies such as glaucoma, neuritis, and multiple sclerosis (MS). In a recent study on 104 MS patients, DTI abnormalities (a drop in FA) were found to be present and correlated with both loss of visual acuity and retinal nerve fiber layer thinning. Thus, DTI can provide functional information of the brain tracts and help detect early changes of the brain before they manifest anatomically. We suggest that DTI abnormalities are likely to be present in individuals subjected to long-duration microgravity, especially those with prolonged exposure to elevated ICP. Using the proposed technology, the project aims to detect changes in the DTI of the brain in individuals that have been exposed to microgravity and to compare them to preflight imaging and with clinical data and other parameters obtained in the process of crewmember monitoring for ICP elevation. This proposal will initiate the first study to evaluate brain microstructures in astronauts beyond anatomical characterization and cerebral spinal fluid (CSF) dynamics, and may offer much improved understanding of the mechanism of ICP elevation and vision changes in a subset of space crewmembers. Quantitative DTI data may also reflect the long-term cumulative effects of multiple microgravity exposures, which is a critical component of knowledge needed to enable human space missions. A 3-Tesla MRI protocol will be used to scan the brain of the study subjects using isovolumetric voxels and 20 diffusion tensor directions. One preflight and one postflight data acquisition will be reviewed to consider the degree of persistence of the quantitative DTI changes in the affected areas of the brain and in optic nerves. These findings will be compared to the astronaut's clinical data and to non-invasive MRI based measurement of intracranial compliance and pressure measurements calculated from the craniospinal CSF fluid dynamics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have demonstrated that there are micro-structural changes in the brains of the astronauts. We still have to further analyze if these changes are reversible or irreversible by analyzing the longitudinal data. Also, our group was the first to identify changes in the macro-structural size of the ventricles before and after returning from space. Direct correlation with the visual impairment syndrome has not been established yet but the changes affect mostly secondary visual pathways. The predominance of the right side in the findings is something we cannot yet explain and will require further follow-up. Advanced images allow to detect changes that cannot be detected with the naked eye and a crucial for the evaluation of the brain of human subjects.
	In the last two years we have further analyzed the information obtained from brain MR images of the 19 crewmembers--3 Female and 16 male, 9 short-duration and 10 long-duration with the age between 40-60 years, consented for retrospective evaluation of their prospectively collected MRIs for this study. The time in flight ranged from 13 to less than 200 days. The goal of this work was to evaluate the effects of microgravity on astronauts' brains using macro- and micro-structural measures by utilizing quantitative magnetic resonance imaging (qMRI), before and after spaceflight. In order to evaluate the structural changes in the brain regions involved in visual function and also any alterations in DTI metrics, including fractional anisotropy (FA) and mean diffusivity (MD), both anatomical (volumetric) and microstructural (diffusion tensor imaging-based) measures were analyzed before and after microgravity exposure. Anatomical MRI were interpolated and reformatted to obtain ~1x1x1 mm transverse volumes that were used by FreeSurfer version 5.3 to provide regional volumes (i.e., lateral ventricular cerebrospinal fluid (CSF), hippocampus, thalamus proper, putamen, caudate, amygdala) and cortical thickness as described elsewhere. Regional cortical thickness values were compared with published literature in adults with age-related changes. One pre-flight and one post-flight data acquisition were used to derive qMRI measures in relevant brain areas including posterior visual cortical gray matter (e.g., peri-calcarine cortex) and white matter fiber pathways (e.g., posterior thalamic radiations). The diffusion-weighted data were decoded using DTIStudio software to obtain the diffusion tensors. The diffusion tensors were then diagonalized to provide the tensor eigenvectors and eigenvalues, which were used to calculate the scalar DTI-derived metrics such as FA and mean, axial, and radial diffusivities. All the operators were blinded to flight status (pre or post) of the astronauts. We used a two-tailed paired t-test to compare both anatomical and DTI measures (pre- vs post-flight) which were showing acceptable normal distribution. Pearson correlation coefficient and linear regression were used to test the association of qMRI measures with age of the astronauts. The results of this investigation are precisely documented in charts, graphs and tables. Below, is a brief summary of our findings along with their possible etiologies:

Task Progress:	I. An overall increase in lateral ventricular volume in the post-flight scans was found. This could be a result of intracranial fluid redistribution, or due to intracranial venous volume reduction caused by relative dehydration in the post-flight scans. Ex-vacuo dilatation caused by deep gray and white matter volume reduction and microanatomy changes might be another possibility, as suggested by decrease in thalamic volume observed in our study. II. Mild thalamic volume reduction with right-sided predominance was observed in our study which could be a result of gray matter changes or rearrangement of the white matter fiber tracts traversing through the thalami including the somatosensory tracts and visual pathways/posterior thalamic radiations. It also might be secondary to proportional inactivity of the astronauts in their space missions or decreased sensorimotor input resulted by microgravity and lack of a large enriched environment. III. The fractional anisotropy was significantly reduced in the right posterior thalamic radiations in post-flight scans. An increase in mean diffusivity in the right occipital cortex, and most significantly, in the right fusiform gyrus, an association region that is linked with various neural pathways related to recognition were also observed in our study. These results may indicate preliminary evidence of cerebral edema and fluid accumulation in the visual pathway which could be secondary to effects of intracranial hypertension induced by microgravity. Alternatively, given previously shown changes of the anterior optic system in microgravity, our findings may be ascribed to possible degeneration of the optic radiation and associate visual cortices secondary to changes of the optic nerve. IV. Volumetric analysis of the cortex demonstrated decreased cortical thickness in the left and right fusiform gyri and also right occipital cortex which were independent of the crewmembers' age. The etiology of this finding needs additional evaluation and correlation with demographic data. Overall, gray and white matter alterations are detected through longitudinal surveillance of astronauts by quantitative MRI before and after spaceflight. Our findings may provide neuroanatomical evidence of brain dysfunction or neuroplasticity in microgravity condition and may offer an improved understanding of the mechanisms involved in intracranial pressure elevation and vision changes in a subset of crew-members. We are currently in the process of designing further researches by utilizing our new technics and expertise to corroborate some of the mechanisms proposed in this study as well as to assess the reversibility of these findings in the astronauts.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Articles in Peer-reviewed Journals	Riascos RF, Kamali A, Hakimelahi R, Mwangi B, Rabiei P, Seidler RD, Behzad BB, Keser Z, Kramer LA, Hasan KM. "Longitudinal analysis of quantitative brain MRI in astronauts following microgravity exposure." J Neuroimaging. 2019 May;29(3):323-30. https://doi.org/10.1111/jon.12609 ; PMID: 30784130 [Note reported originally in Feb 2018 as "Submitted to Lancet-Neurology and under review as of January 2018."] , May-2019
Articles in Peer-reviewed Journals	Hasan KM, Mwangi B, Keser Z, Riascos R, Sargsyan AE, Kramer LA. "Brain quantitative MRI metrics in astronauts as a unique professional group." J Neuroimaging. 2018 May;28(3):256-68. https://doi.org/10.1111/jon.12501 ; PMID: 29460455 , May-2018